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ASSIGNMENT #5

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
Professor: Dr. Michael Caracotsios
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Your name: JAKE PROCTOR 679279362
Due Date: Saturday April 04 by 12:00 noon

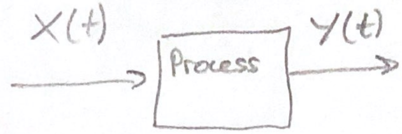
Number of points........................100
Your score..............................

INSTRUCTIONS FOR ASSIGNMENT DELIVERY

- Scan your Assignment and create a PDF file
- PDF File Name: Assignment5_XY.PDF
  - **X**: First letter of your first name
  - **Y**: Your last name
  - Example: Assignment5_MCaracotsios.PDF
- Email your PDF file to: Daniel Christiansen dchris28@uic.edu and Yuechen Gao ygao54@uic.edu by the due date (Saturday April 04) and time (12:00 noon)

Note: **All problems are worth 100 points. You final score will be the average value**

PROBLEM 5a:



CV: tank pressure
MV: Valve opening on valve 1 (flowrate F_1)

Balance: Accum = IN − OUT

$$\frac{dn}{dt} = F_1 - F_2 \quad ; \quad n \equiv \text{total mols} \quad \text{but we know for ideal gas } n = \frac{PV}{RT}$$

$$\frac{d\left[\frac{PV}{RT}\right]}{dt} = F_1 - F_2 \Rightarrow \frac{V}{RT}\frac{dP}{dt} = X_1 C_{v_1}\sqrt{P_1 - P} - X_2 C_{v_2}\sqrt{P - P_2} \qquad \text{eqn 1}$$

@ $t=0$ $\quad P = P^{ss}, \; X_1 = X_1^{ss}$

$$0 = X_1^{ss} C_{v_1}\sqrt{P_1 - P^{ss}} - X_2 C_{v_2}\sqrt{P^{ss} - P_2} \qquad \text{eqn 2}$$

Set $X = X_1 - X_1^{ss}$ and $Y = P - P^{ss}$

subtract eqn 2 from eqn 1

$$\frac{V}{RT}\frac{dP}{dt} = X_1 C_{v_1}\sqrt{P_1 - P} - X_2 C_{v_2}\sqrt{P - P_2} - X_1^{ss} C_{v_1}\sqrt{P_1 - P^{ss}} + X_2 C_{v_2}\sqrt{P^{ss} - P_2}$$

now need to linearize terms (linearize about P^{ss})

$$\therefore X_1 C_{v_1}\sqrt{P_1 - P} \approx \cancel{X_1^{ss} C_{v_1}\sqrt{P_1 - P^{ss}}} - X_1^{ss} C_{v_1}\left(\tfrac{1}{2}\right)\frac{(P - P^{ss})}{\sqrt{P_1 - P^{ss}}} + C_{v_1}(X_1 - \cancel{X_1^{ss}})\sqrt{P_1 - P^{ss}}$$

and

$$\sqrt{P - P_2} \approx \sqrt{P^{ss} - P_2} + \frac{1}{2}\frac{P - P^{ss}}{\sqrt{P^{ss} - P_2}}$$

$\therefore$ sub those linearizations

$$\frac{V}{RT}\frac{dP}{dt} = \frac{-X_1^{ss} C_{v_1}(P - P^{ss})}{2\sqrt{P_1 - P^{ss}}} + X_1 C_{v_1}\sqrt{P_1 - P^{ss}} - X_2 C_{v_2}\left[\sqrt{P^{ss} - P_2} + \frac{P - P^{ss}}{2\sqrt{P^{ss} - P_2}}\right] \ldots$$

$$\ldots - X_1^{ss} C_{v_1}\sqrt{P_1 - P^{ss}} + X_2 C_{v_2}\sqrt{P^{ss} - P_2}$$

expand out

$$\frac{V}{RT}\frac{dY}{dt} = X_1 C_{v_1}\sqrt{P_1-P^{ss}} - X_1^{ss}C_{v_1}\left(\frac{Y}{2\sqrt{P_1-P^{ss}}}\right) - \cancel{X_2 C_{v_2}\sqrt{P^{ss}-P_2}} - X_2 C_{v_2}\left(\frac{Y}{2\sqrt{P^{ss}-P_2}}\right) \ldots$$

$$\ldots - X_1^{ss}C_{v_1}\sqrt{P_1-P^{ss}} + \cancel{X_2 C_{v_2}\sqrt{P^{ss}-P_2}}$$

Simplify

$$X_1 C_{v_1}\sqrt{P_1-P^{ss}} - X_1^{ss}C_{v_1}\sqrt{P_1-P^{ss}} \Rightarrow C_{v_1}\sqrt{P_1-P^{ss}}\left(X_1 - X_1^{ss}\right) = \underline{C_{v_1}\sqrt{P_1-P^{ss}}\,X}$$

$$-X_1^{ss}C_{v_1}\left(\frac{Y}{2\sqrt{P_1-P^{ss}}}\right) - X_2 C_{v_2}\left(\frac{Y}{2\sqrt{P^{ss}-P_1}}\right) \Rightarrow -\left(\frac{X_1^{ss}C_{v_1}}{2\sqrt{P_1-P^{ss}}} + \frac{X_2 C_{v_2}}{2\sqrt{P^{ss}-P_2}}\right)Y$$

∴ we have ...

$$\frac{V}{RT}\frac{dY}{dt} = C_{v_1}\sqrt{P_1-P^{ss}}\,X - \left(\frac{X_1^{ss}C_{v_1}}{2\sqrt{P_1-P^{ss}}} + \frac{X_2 C_{v_2}}{2\sqrt{P^{ss}-P_2}}\right)Y$$

we know $G(s) = \frac{Y}{X} \Rightarrow \frac{K_p}{\tau_p s + 1}$

need the eqn of the form: $\tau_p \frac{dY}{dt} + Y = K_p X$

$$\Rightarrow \left[\frac{\frac{V}{RT}}{\frac{X_1^{ss}C_{v_1}}{2\sqrt{P_1-P^{ss}}} + \frac{X_2 C_{v_2}}{2\sqrt{P^{ss}-P_2}}}\right]\frac{dY}{dt} + Y = \left[\frac{C_{v_1}\sqrt{P_1-P^{ss}}}{\frac{X_1^{ss}C_{v_1}}{2\sqrt{P_1-P^{ss}}} + \frac{X_2 C_{v_2}}{2\sqrt{P^{ss}-P_2}}}\right]X$$

$$\therefore \boxed{K_p = \frac{C_{v_1}\sqrt{P_1-P^{ss}}}{\frac{X_1^{ss}C_{v_1}}{2\sqrt{P_1-P^{ss}}} + \frac{X_2 C_{v_2}}{2\sqrt{P^{ss}-P_2}}}} \quad \text{and} \quad \boxed{\tau_p = \frac{\frac{V}{RT}}{\frac{X_1^{ss}C_{v_1}}{2\sqrt{P_1-P^{ss}}} + \frac{X_2 C_{v_2}}{2\sqrt{P^{ss}-P_2}}}}$$

PROBLEM 5b :

Forms of models

First order (FO) : $\frac{Y(s)}{X(s)} = \frac{K_p}{\tau_p s + 1} \Rightarrow \tau_p s Y(s) + Y(s) = K_p X(s)$

w/ differential eqn $\tau_p \frac{dY}{dt} = -Y + K_p X(t)$

FOPTD : $\frac{Y(s)}{X(s)} = \frac{K_p e^{-s\theta_p}}{\tau_p s + 1} \Rightarrow \tau_p s Y(s) + Y(s) = K_p X(s) e^{-s\theta_p}$

w/ Dif eqn: $\tau_p \frac{dY}{dt} = -Y + K_p X_d(t - \theta_p)$

SO : $\frac{Y(s)}{X(s)} = \frac{K_p}{\alpha_2 s^2 + \alpha_1 s + \alpha_0}$

w/ dif eqns: $U_1 = \frac{dY}{dt} \quad \| \quad \alpha_2 \frac{dU_1}{dt} = -\alpha_1 U_1 - \alpha_0 U_2 + K_p X(t)$

$U_2 = Y \quad \| \quad \frac{dU_2}{dt} = U_1$

SOPTD: $\frac{Y(s)}{X(s)} = \frac{K_p e^{-s\theta_p}}{\alpha_2 s^2 + \alpha_1 s + \alpha_0}$

w/ dif eqns: $U_1 = \frac{dY}{dt} \quad \| \quad \alpha_2 \frac{dU_1}{dt} = -\alpha_1 U_1 - \alpha_0 U_2 + K_p X_d(t - \theta_p)$

$U_2 = Y \quad \| \quad \frac{dU_2}{dt} = U_1$

Working forms of the functions (solved) → Δ in CV → Steady State value

FO: $Y(t) = K_p\left(1-e^{-t/\tau_p}\right)(-50) + 50$

FOPTD: $Y(t) = K_p\left(1-e^{-(t-\theta)/\tau_p}\right)(-50) + 50$

SO: $Y(t) = K_p\left[1 + \frac{1}{\tau_2-\tau_1}\left(\tau_1 e^{-t/\tau_1} - \tau_2 e^{-t/\tau_2}\right)\right](-50) + 50$

SOPTD: $Y(t) = K_p\left[1 + \frac{1}{\tau_2-\tau_1}\left(\tau_1 e^{-(t-\theta)/\tau_1} - \tau_2 e^{-(t-\theta)/\tau_2}\right)\right](-50) + 50$

Notice in the raw data there appears to be both pure and apparent dead time

~~Pure dead time = 20~~

$\cancel{S(t-\theta) = \begin{cases} 0 & \text{for dead time} < 20 \\ 20 & \text{for dead time} > 20 \end{cases}}$

each model was plugged into Excel and solver was used to peform non-linear regression

* see table attached showing all values of interest.

Note the postulated models are meant to lead to a negative value for K_p because the step was down therefore $K_p < 0$

(attached is also a screenshot of the excel sheet set up for calculations)

Analysation of AIC values suggests the <u>best model</u> to fit the raw data is $\boxed{\text{FOPTD}}$

Table 1: Results summary for Problem 5b

Model	Kp (NH3 ppm / gpm)	Tau1 (s)	Tau2 (s)	Theta (s)	RSQ	AIC
1	-0.051	155.908	N/A	N/A	0.429	-59.565
2	-0.038	64.326	N/A	27.243	0.028	-105.608
3	-0.039	44.914	44.914	N/A	0.091	-84.025
4	-0.037	14.754	54.458	17.251	0.024	-104.458

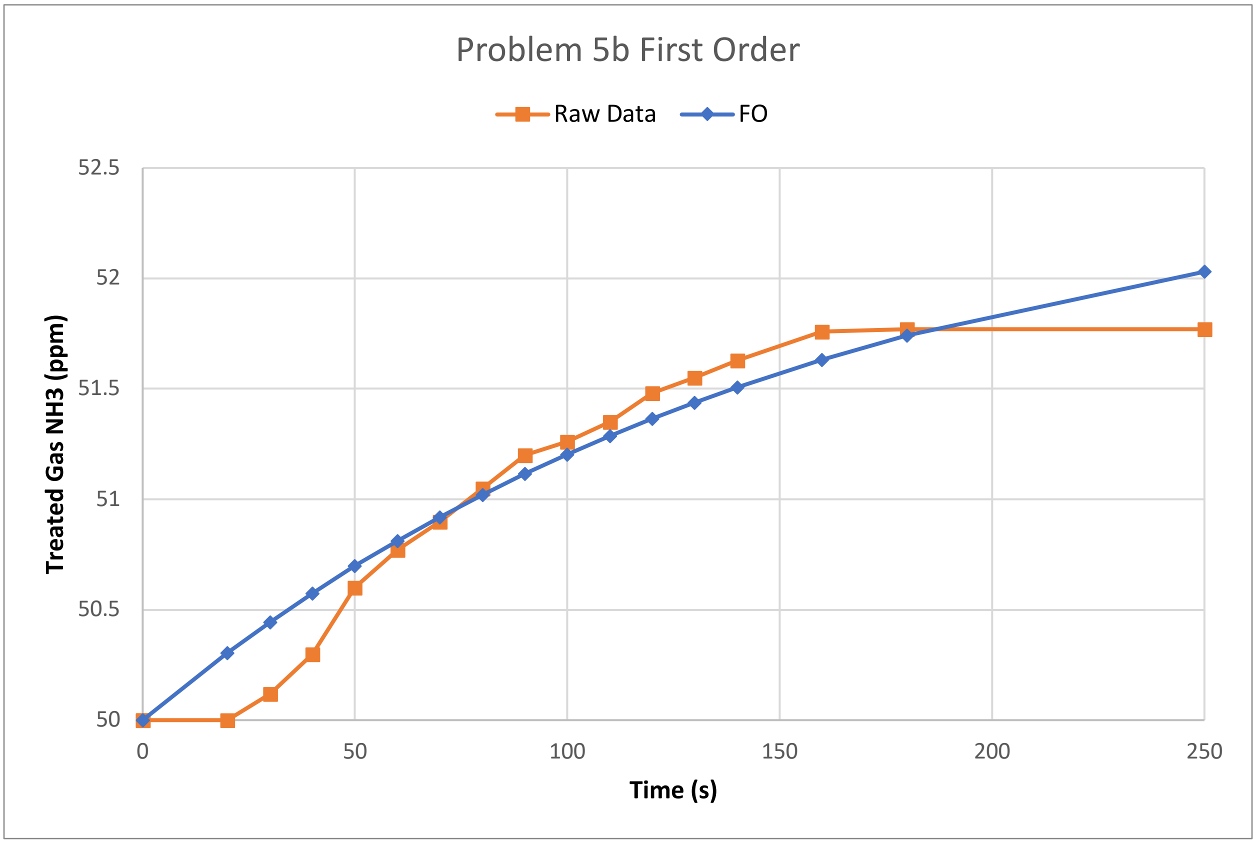


Figure 1: First order (FO) model fit

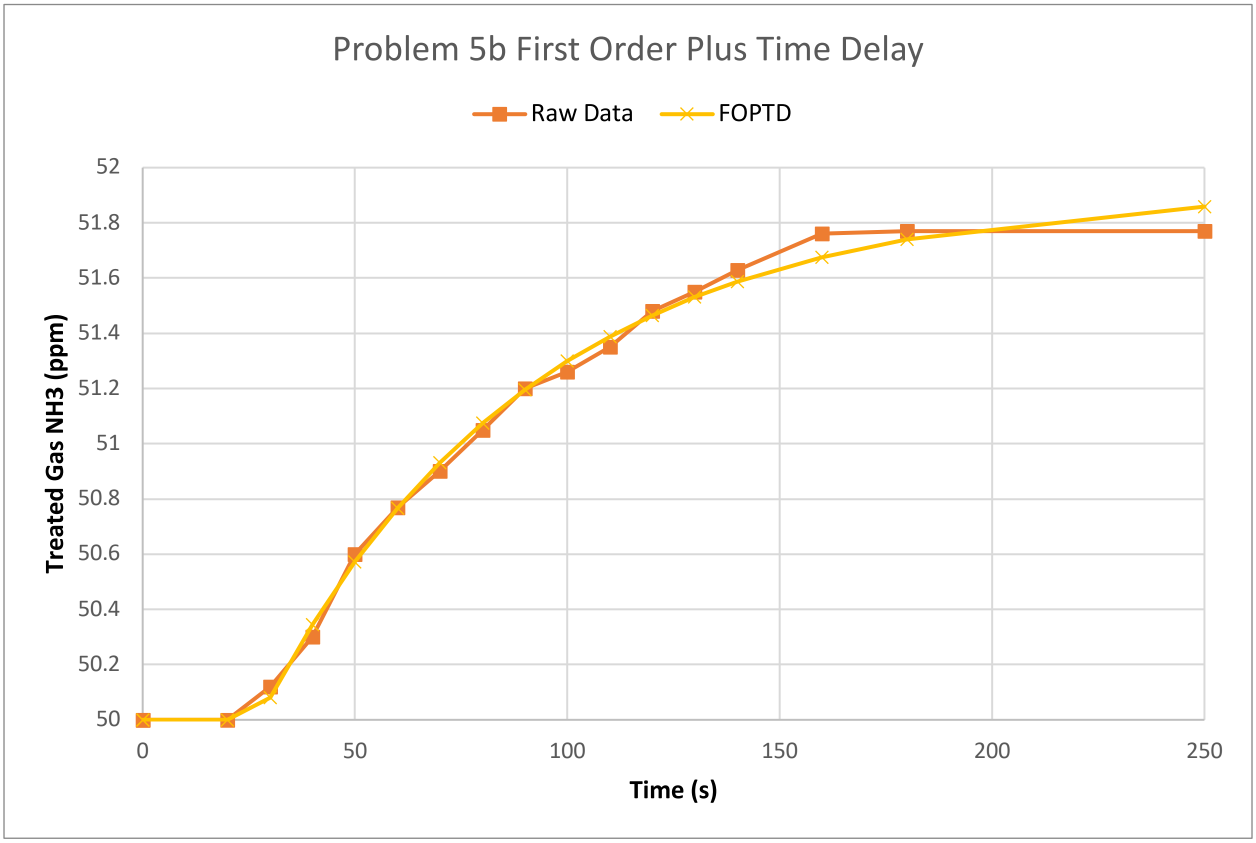


Figure 2: First order plus time delay (FOPTD) model fit

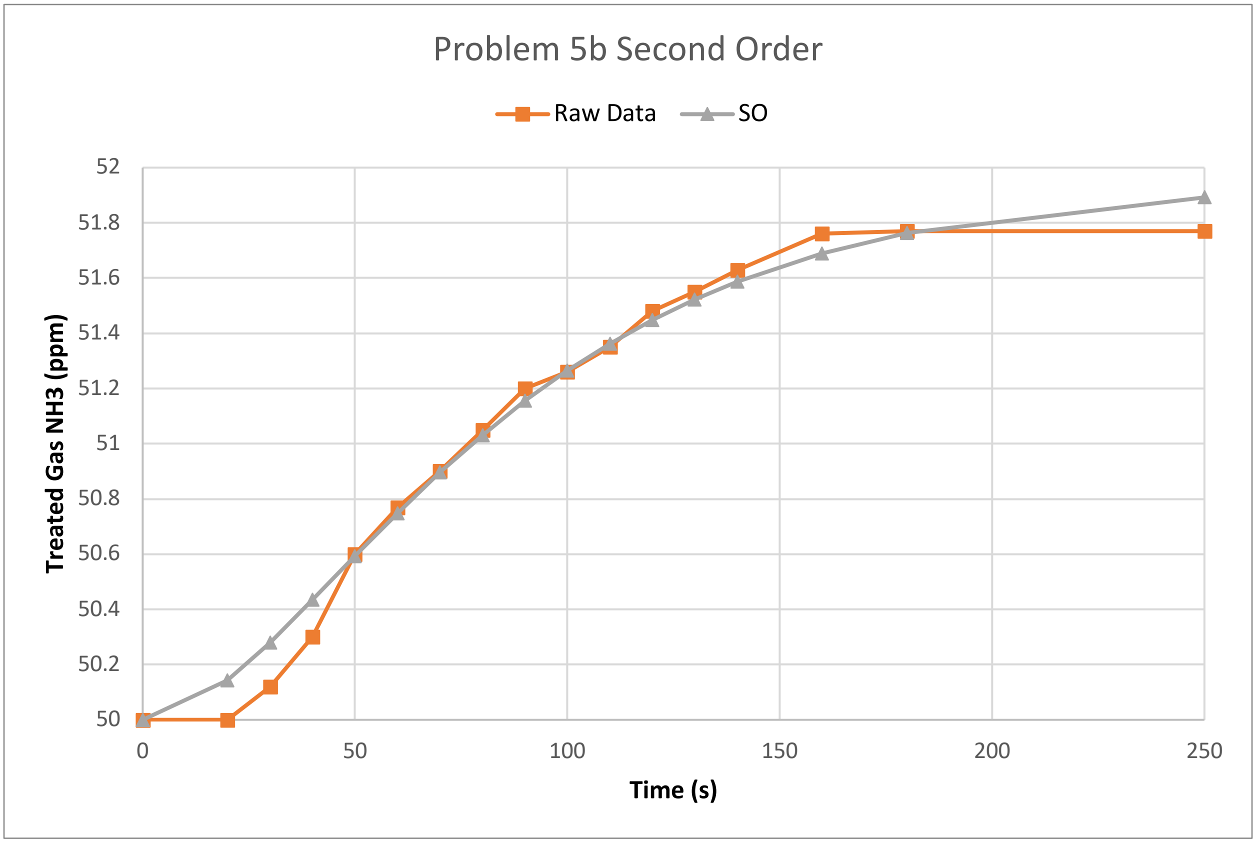


Figure 3: Second order (SO) model fit

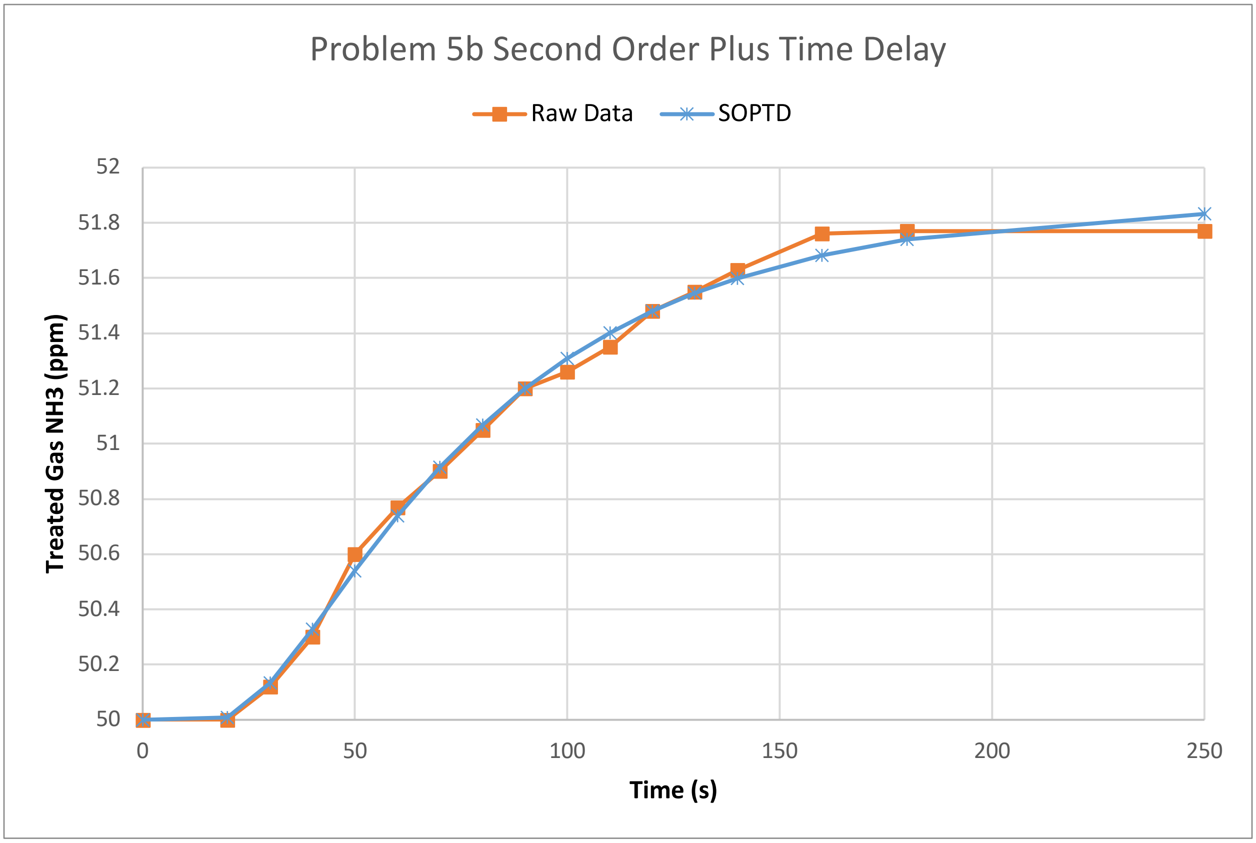


Figure 4: Second order plus time delay (SOPTD) model fit

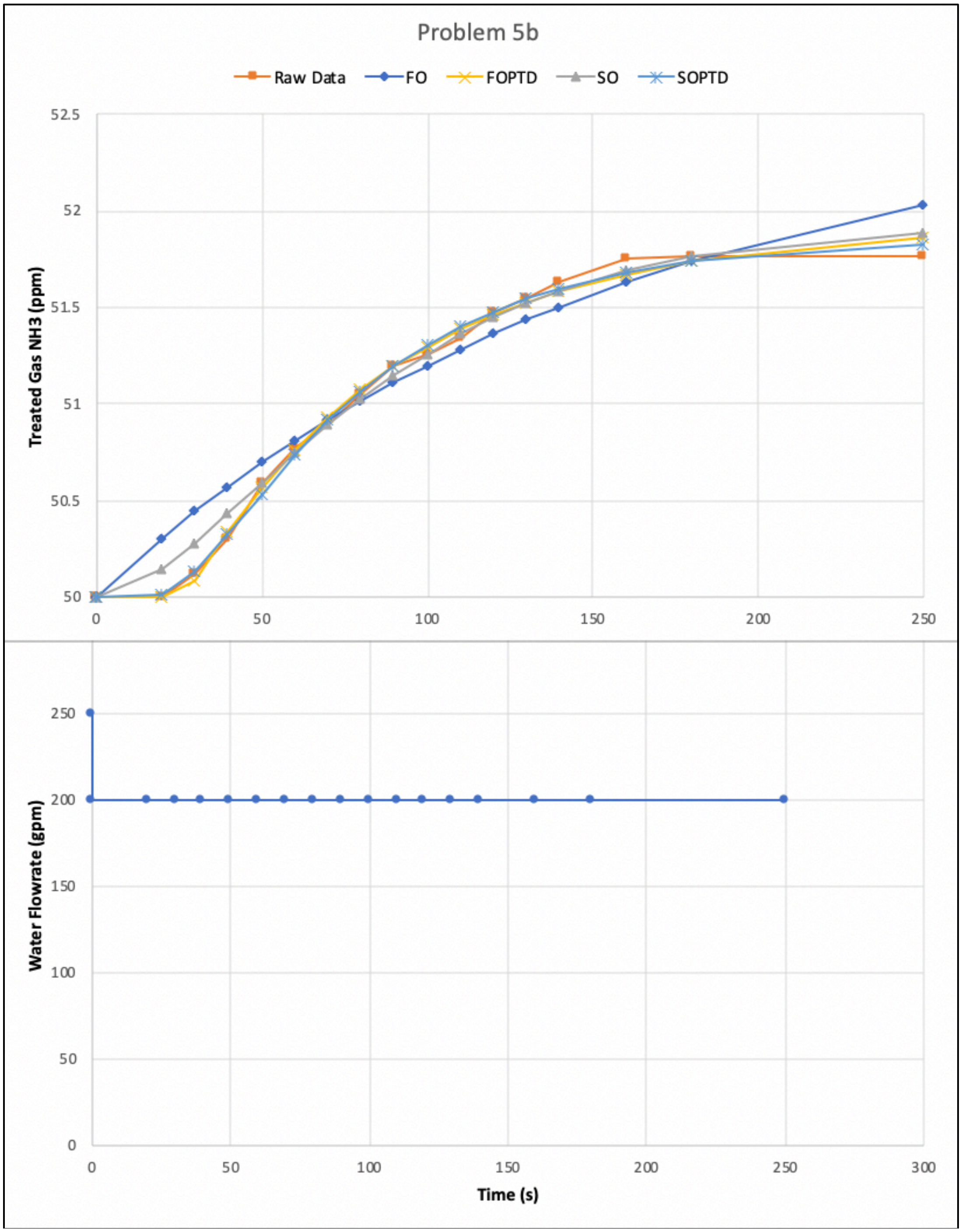


Figure 5: Model comparison against raw data

Time, t (s)	Water Flow (gpm)	Treated Gas NH3 (pp	FO (ppm)	residuals ^2	FOPTD (ppm)	residuals ^2	SO (ppm)	residuals ^2	SOPTD (ppm	residuals ^2
0	250	50	50	0	50	0	50	0	50	0
0	200	50	50	0	50	0	50	0	50	0
20	200	50	50.30597229	0.093619	50	0	50.14380476	0.02067981	50.0081262	6.6036E-05
30	200	50.12	50.44485457	0.1055305	50.08051723	0.001558889	50.28092435	0.02589665	50.1331781	0.00017366
40	200	50.3	50.57510891	0.0756849	50.34521178	0.002044105	50.43490256	0.0181987	50.3292886	0.00085782
50	200	50.6	50.6972713	0.0094617	50.57179627	0.00079545	50.59354636	4.165E-05	50.5392303	0.00369296
60	200	50.77	50.81184446	0.001751	50.7657577	1.79971E-05	50.74885854	0.00044696	50.7376793	0.00104463
70	200	50.9	50.91929985	0.0003725	50.93179305	0.001010798	50.89584855	1.7235E-05	50.9148839	0.00022153
80	200	51.05	51.02007967	0.0008952	51.07392306	0.000572313	51.03165083	0.00033669	51.0684779	0.00034143
90	200	51.2	51.11459863	0.0072934	51.19558954	1.94522E-05	51.15487532	0.00203624	51.199411	3.4687E-07
100	200	51.26	51.20324567	0.0032211	51.29973879	0.001579171	51.26513262	2.6344E-05	51.3099562	0.00249562
110	200	51.35	51.2863856	0.0040468	51.38889289	0.001512657	51.36268989	0.00016103	51.4027573	0.00278333
120	200	51.48	51.36436052	0.0133725	51.46521081	0.00021872	51.44822427	0.0010097	51.480397	1.5759E-07
130	200	51.55	51.43749132	0.0126582	51.53054067	0.000378666	51.52264857	0.0007481	51.5452185	2.2862E-05
140	200	51.63	51.50607892	0.0153564	51.5864645	0.00189534	51.58699018	0.00184984	51.5992707	0.00094429
160	200	51.76	51.630736	0.0167092	51.67531612	0.00717136	51.6896436	0.00495002	51.681818	0.00611243
180	200	51.77	51.74038575	0.000877	51.74042433	0.000874721	51.76421901	3.342E-05	51.7390473	0.00095807
250	200	51.77	52.0301691	0.067688	51.85882673	0.007890187	51.89212989	0.01491571	51.8323875	0.0038922

	FO	FOPTD	SO	SOPTD
Kp	-0.050830641	-0.038379227	-0.038818	-0.037361407
tau1	155.9145145	64.32581128	44.914241	54.45755398
tau2	NA	0	44.91414	14.75354488
theta	NA	27.24270959	NA	17.250744
SUM res	0.42853733	0.027539825	0.0913481	0.023607373
k (# of parameters)	2	3	3	4
n	18	18	18	18

Model	Kp	Tau1	Tau2	Theta	RSQ	AIC
1	-0.050830641	155.9145145	N/A	N/A	0.42853733	-59.5651996
2	-0.038379227	64.32581128	N/A	27.24270959	0.027539825	-105.607967
3	-0.03881755	44.9142407	44.91414	N/A	0.091348104	-84.0251681
4	-0.037361407	54.45755398	14.753545	17.2507436	0.023607373	-104.458223